

Henry Boyes

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Data & ML engineer with production experience building end-to-end pipelines with HPC orchestration, ETL, deep learning inference, and deployed data tools at clinical scale. Brings operational discipline from the US Navy operations environment, where **data accuracy** and **process reliability** are mission critical.

TECHNICAL SKILLS

Languages & ML: Python | R | SQL | Bash

Libraries/Frameworks: Pandas | NumPy | scikit-learn | statsmodels | matplotlib | seaborn | Plotly | Streamlit | CustomTkinter | openpyxl | BeautifulSoup | SQLAlchemy | Cellpose | Scanpy | Squidpy

ML & Modeling: Supervised/unsupervised learning | deep learning inference | clustering (K-means, hierarchical, Leiden/UMAP) | PCA | spatial statistics | feature engineering

Tools: Git | Docker | Docker Hub | GitHub Actions | Slurm | ETL workflows | GitHub Copilot | Claude Code

Databases: PostgreSQL | SQLite | MySQL

PROFESSIONAL EXPERIENCE

Research Technician II – Cleveland Clinic Foundation

Mar 2025 – Present

- Designed and deployed a **containerized production spatial proteomics ML pipeline** processing ~3M cells and automated image builds via **GitHub Actions CI/CD**, **Docker**, GPU segmentation (Cellpose v4), Leiden clustering/UMAP, spatial analysis (Squidpy). github.com/boyeshenry-byte/akoya-pcf-pipeline
- Engineered **Slurm HPC orchestration** across GPU (A100) and CPU nodes, fully reproducible, minimal manual intervention.
- Built **Python ETL workflows cutting data prep 40%**; SQLite job tracking and centralized logging for pipeline observability and failure diagnosis.
- Replaced 47-sheet Excel log with **full-stack desktop app** (Python/CustomTkinter/SQLite), multi-user, CRUD, filtered export, soft delete. github.com/boyeshenry-byte/lab-accession-app
- Identified CCL19 as the sole statistically stable response predictor for a 48-protein cytokine panel (n=21) via **elastic net regularization** and **500-iteration bootstrap stability selection**; **cross-validated AUC=0.68**, directionally confirmed by **ordinal regression**.
- Managed **800+ oncology patient records**; enforced data quality, integrity, and HIPAA compliance across cross-functional teams.

Research Assistant – Kent State University

Aug 2023 – Oct 2024

- Contributed statistical analysis to a **peer-reviewed publication** in Acta Biomaterialia (Aug 2025); standardized analytical practices across the team.

HM2, Quality Management – U.S. Navy, NMCS D

Dec 2018 – Apr 2020

- Led data modernization migrating patient information from Excel to a structured digital platform, improving **HEDIS metrics +60%** and training readiness **60%→95% in one month** via data-driven process redesign.
- Supervised 2 staff (both promoted); awarded **Navy Achievement Medal** and **Dept. Sailor of the Year**.

HM3 / BM3 – U.S. Navy, NMCS D & USS Porter (DDG 78)

Dec 2012 – Dec 2018

- Floor corpsman in Med/Surg ICU; maintained patient records and trained staff on documentation and compliance.
- Awarded **Arch Griffith Leadership Award**, **Blue Jacket of the Year** for demonstrated leadership.

PROJECTS

Job Market Analyzer | Python, BeautifulSoup, SQLite, SQLAlchemy, PostgreSQL, GitHub Actions, Streamlit, Plotly | github.com/boyeshenry-byte/job_market_analyzer

- Built automated multi-source scraping pipeline (3 platforms) with **GitHub Actions CI/CD scheduling**, SQLAlchemy-backed **PostgreSQL** storage and deployed **Streamlit/Plotly dashboard** for trend analyzation. Live at <https://job-market-analyzer.streamlit.app>

EDUCATION & CERTIFICATIONS

University of Pittsburgh (Expected 2027) | M.S. Data Science | Coursework: Machine Learning, Statistical Analysis, Database Systems

Kent State University (2024) | B.S. Organismal Biology, cum laude (3.6 GPA) | A.S. Biology (3.4 GPA) |

Departmental Award for Significant Academic Achievement

Lean Six Sigma Yellow Belt (2019) | Project Management Performance Excellence (2019) | IBM AI Fundamentals (2025) | IBM Data Fundamentals (2025)

Previously held U.S. Secret Security Clearance